NoisePrivacyMap<L1Distance<RBig>, MaxDivergence> for ZExpFamily<1>

Michael Shoemate

This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of `NoisePrivacyMap` for `ZExpFamily<1>` in `mod.rs` at commit f5bb719 (outdated[1]).

1 Hoare Triple

Precondition

Compiler-Verified

`NoisePrivacyMap` is parameterized as follows:

- `MI`, the input metric, is of type `L1Distance<RBig>`
- `MO`, the output measure, is of type `MaxDivergence`

User-Verified

None

Pseudocode

```
# analogous to impl NoisePrivacyMap<L1Distance<RBig>, MaxDivergence> for ZExpFamily<1> in Rust
class ZExpFamily1:
    def noise_privacy_map(
        self, _input_metric: L1Distance[RBig], _output_measure: MaxDivergence
    ) -> PrivacyMap[L1Distance[RBig], MaxDivergence]:
        scale = self.scale
        if scale < RBig.ZERO:  #
            raise "scale must be non-negative"

        def privacy_map(d_in: RBig):
            if d_in < RBig.ZERO:  #
                raise "sensitivity must be non-negative"

            if d_in.is_zero():  #
                return 0.0

            if scale.is_zero():  #
                return float("inf")

```

[1] See new changes with `git diff f5bb719..7a359e80 rust/src/measurements/noise/distribution/laplace/mod.rs`

```
20            return f64.inf_cast(d_in / scale)  #
21
22        return PrivacyMap.new_fallible(privacy_map)
```

Postcondition

Theorem 1.1. Given a distribution `self`, returns `Err(e)` if `self` is not a valid distribution. Otherwise if the output is `Ok(privacy_map)` then `privacy_map` observes the following:

Define `function(x) = x + Z` where `Z` is a vector of iid samples from `self`.

For every pair of elements x, x' in `VectorDomain<AtomDomain<IBig>>`, and for every pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for `output_measure`, if x, x' are `d_in`-close under `input_metric`, `privacy_map(d_in)` does not raise an exception, and `privacy_map(d_in)` $\leq$ `d_out`, then `function(x)`, `function(x')` are `d_out`-close under `output_measure`.

Proof. Line 7 rejects `self` if `self` does not represent a valid distribution, satisfying the error conditions of the postcondition.

We now construct the privacy map. First consider the extreme values of the scale and sensitivity parameters. The sensitivity `d_in`, a bound on distances, must not be negative, as checked on line 11. In the case where sensitivity is zero (line 14), the privacy loss is zero, regardless the choice of scale parameter (even zero). This is because the privacy loss when adjacent datasets are always identical is zero. Otherwise, in the case where the scale is zero, the privacy loss is infinite. To avoid a rational division overflow, line 17 returns infinity.

By line 20, both the sensitivity and scale are positive rationals. We now show that the same privacy bound holds in both the non-modular and modular cases.

If the metric is non-modular, the standard discrete-Laplace argument gives

$$\max_{x \sim x'} D_\infty(M(x), M(x')) \leq \frac{\texttt{d_in}}{s},$$

where s denotes `self.scale`.

Suppose now that the metric is modular. Let

$$G = \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_d}$$

denote the quotient group induced by the coordinatewise moduli, and let $\pi : \mathbb{Z}^d \to G$ be coordinatewise reduction modulo those moduli. The mechanism implemented by `self` is the wrapped additive mechanism

$$M_m(x) = \pi(x + Y),$$

where Y has iid discrete-Laplace coordinates with scale s.

Consider neighboring inputs $x \sim x'$ and let $r = x' - x \in G$. By the definition of modular L_1 distance, there exists a lift $t \in \mathbb{Z}^d$ such that $\pi(t) = r$ and $\|t\|_1 \leq$ `d_in`. For any output residue $u \in G$, choose an integer representative $z \in \mathbb{Z}^d$ of $u - x$. Then

$$\Pr[M_m(x) = u] = \frac{1}{Z_s} \sum_{k \in \mathbb{Z}^d} \exp\left(-\frac{\|z + Mk\|_1}{s}\right),$$

and

$$\Pr[M_m(x') = u] = \frac{1}{Z_s} \sum_{k \in \mathbb{Z}^d} \exp\left(-\frac{\|z - t + Mk\|_1}{s}\right),$$

where M is the diagonal matrix of coordinatewise moduli.

For every $k \in \mathbb{Z}^d$, the triangle inequality gives

$$\|z + Mk\|_1 = \|(z - t + Mk) + t\|_1 \geq \|z - t + Mk\|_1 - \|t\|_1.$$

Therefore

$$\exp\left(-\frac{\|z+Mk\|_1}{s}\right) \le e^{\|t\|_1/s} \exp\left(-\frac{\|z-t+Mk\|_1}{s}\right).$$

Summing over k yields

$$\Pr[M_m(x)=u] \le e^{\|t\|_1/s} \Pr[M_m(x')=u] \le e^{\texttt{d_in}/s} \Pr[M_m(x')=u].$$

Since this pointwise likelihood-ratio bound holds for every output residue, it also holds for every measurable subset of G, so

$$D_\infty(M_m(x), M_m(x')) \le \frac{\texttt{d_in}}{s}.$$

Line 20 implements this bound with exact division and conservative cast to float. □